THE CURIOUS ELECTRIC CO.

# Solar Bauble Instructions

| Date: | 22/10/2026 | Version: | 1.0 | By: | Matt Little |
|---|---|---|---|---|---|





The Solar Bauble kit is a great geeky way to celebrate the festive period! The LEDs flash for a while in the evening and the unit recharges during the day.

It is a DIY soldering kit, with relatively easy soldering required.

A small solar panel charges some capacitors during the day. Then at night the unit will start to randomly twinkle, with eight red and green LEDs around the circuit board. Firmware on the microcontroller has been designed to be very low power, and, with fully charged capacitors, the LEDs should flash for up to 2 hours.

It's designed as a fun and interactive way introduce solar power and improve soldering skills.

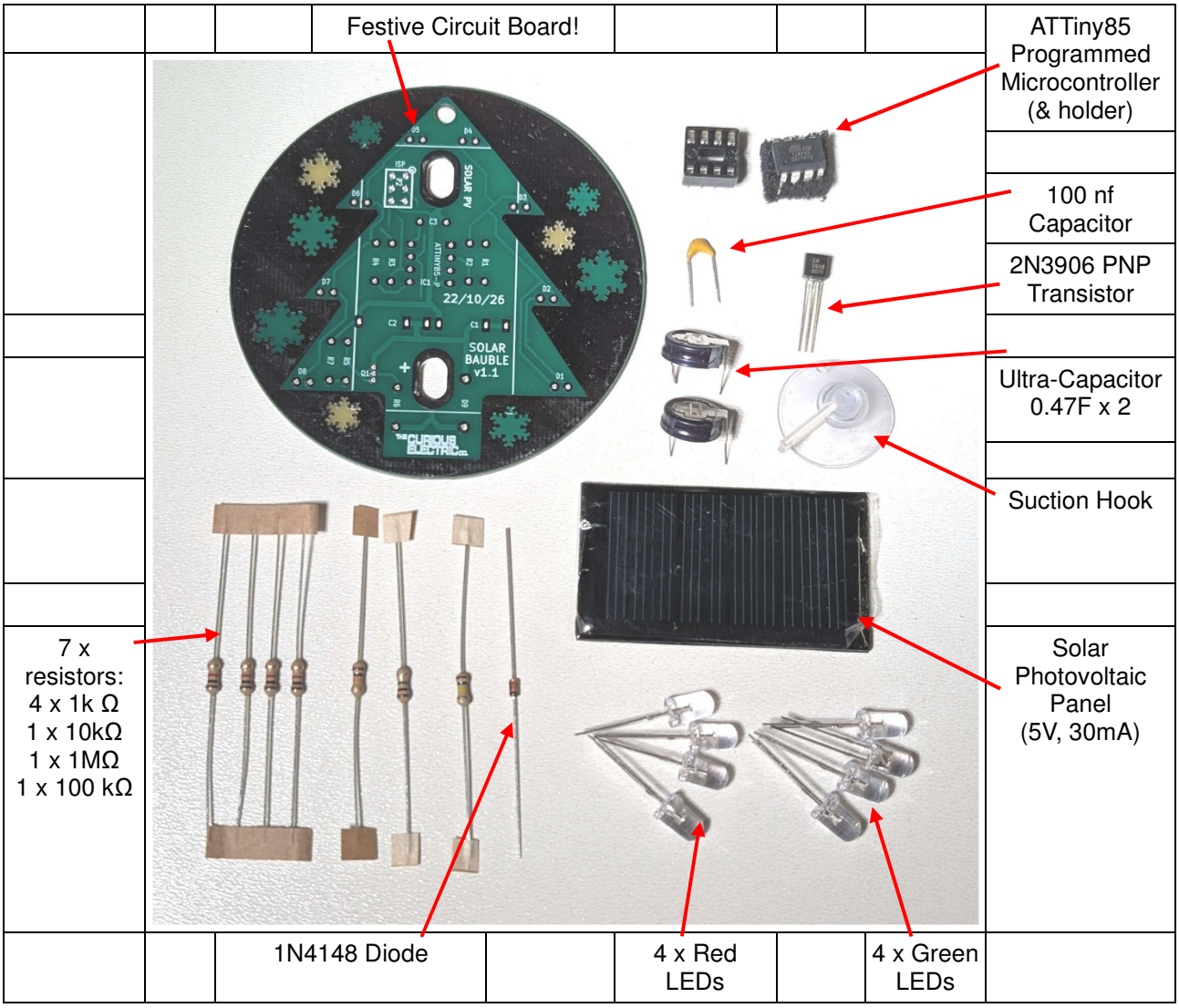
It does need to be placed on a SUNNY window. Preferably south facing or that gets direct sun for some of the day. (It will not charge well from indoor lighting). It will need at least 1-2 hours of sunlight to fully charge.

***Note: Only suitable for people over the age of 8 years old, due to small parts.***

The firmware is open and available to change as you might want for different patterns etc. Please see the GitHub repository here:

**https://github.com/curiouselectric/solarBauble**

## Parts included:

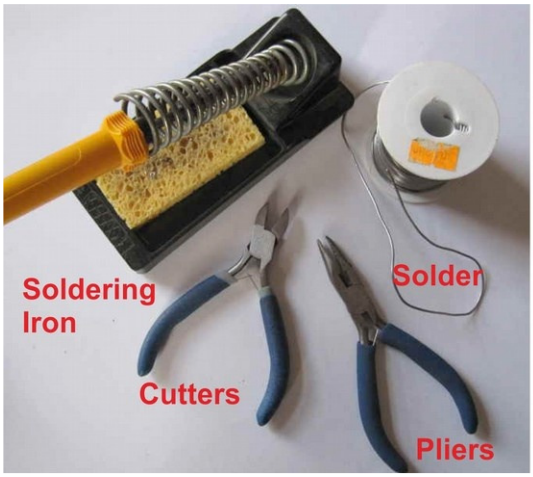



## Parts list:

| Item | Ref | Item | Ref |
|---|---|---|---|
| 0.47F Ultra Capacitor | C1, C2 | Printed Circuit Board | PCB |
| 100nF Decoupling Capacitor | C3 | 1k Ω Resistor | R1 – R4 |
| 5mm Red LEDs (loose) | D1, D3, D5, D7 | 100 kΩ Resistor | R5 |
| 5mm Green LEDs (in a bag) | D2, D4, D6, D8 | 1 MΩ Resistor | R6 |
| 1N4148 Diode | D9 | 10 kΩ Resistor | R7 |
| ATTiny85 Microcontroller (programmed) + 8 pin IC Holder | IC1 | Solar Photovoltaic Panel (5V, 30mA) | SC1 |
| Suction Cup Hook (to hold PCB to glass window) | N/A | Transistor 2N3906 PNP | Q1 |

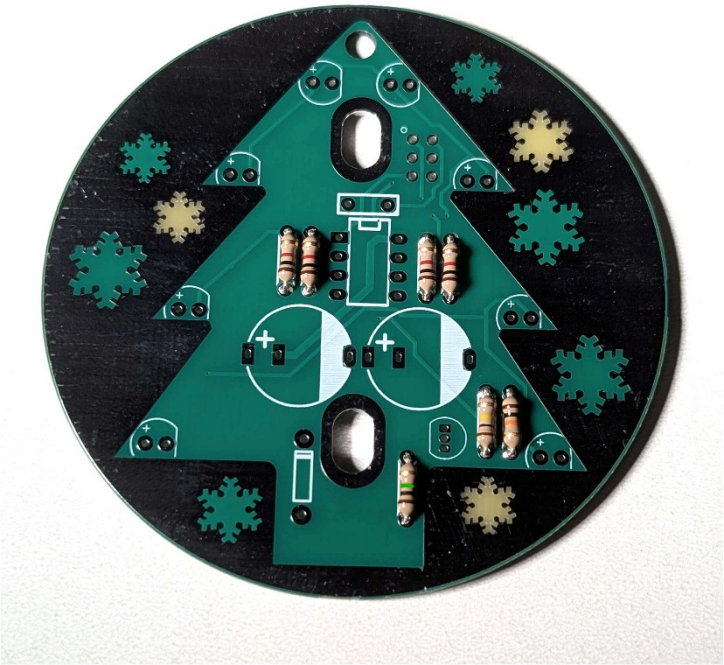
## Tools required:

Basic soldering tools are required, including a soldering iron, lead-free solder and some good wire cutters.



PCB Instructions:

| **Step: 1** | Solder resistors |
|---|---|



| Value | Ref | Colour |
|---|---|---|
| 120 Ω | R1 – R4 | Brown – Red – Brown – Gold |
| 100 kΩ | R5 | Brown – Black – Yellow - Gold |
| 1 MΩ | R6 | Brown – Black – Green - Gold |
| 10 kΩ | R7 | Brown – Black – Orange - Gold |

Insert and solder the resistors to the locations marked on the PCB.

Polarity does not matter for resistors. Ensure they are soldered flat & close to the PCB.

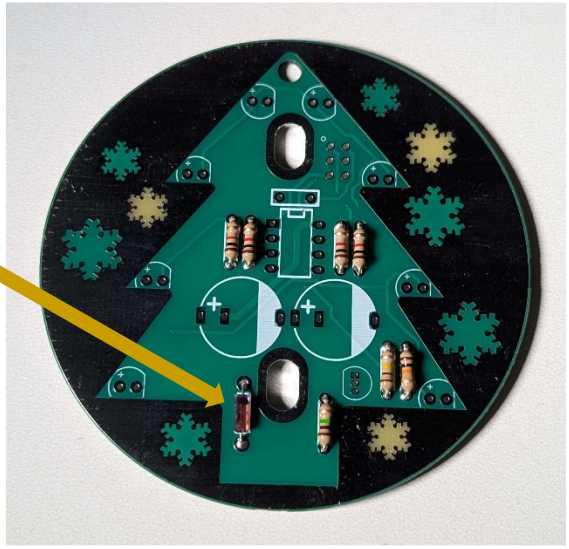
You can use a multi-meter or colour code chart to ensure correct resistor values.

| **Step: 2** | Solder diode (D9) |
|---|---|

Ensure correct orientation for the 1N4148 diode component, D9.

This has tiny markings saying “1N4148” and a black band around one end to indicate polarity.

Ensure the black band on the diode aligns with the white band marked on the PCB.

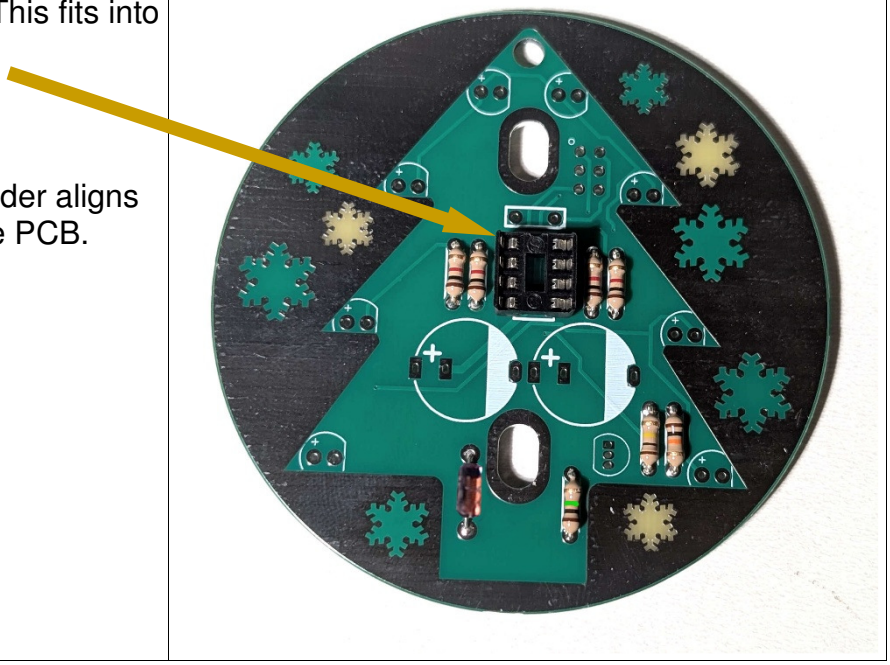


| **Step: 3** | Solder IC holders |
|---|---|

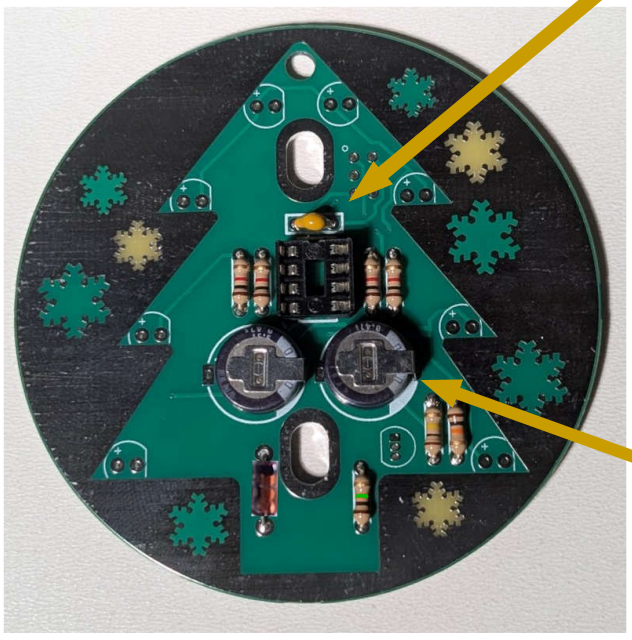
Solder the 8-pin IC holder. This fits into the areas marked IC1

Ensure correct orientation!

Ensure the notch on the holder aligns with the notch shown on the PCB.



| **Step: 4** | Solder capacitors |
|---|---|

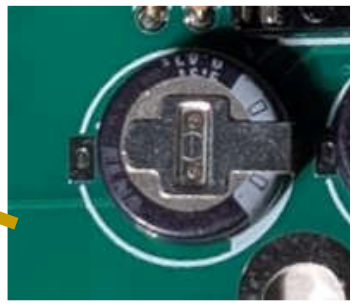


First solder the non-polarised capacitor, C2. Orientation of these does not matter:

| Value | Ref | Marking |
|---|---|---|
| 100nf | C2 | 104 |

Next solder the polarised ultra capacitor. Ensure correct orientation of this component.

The negative lead is marked with a white print on the capacitor. The negative lead comes from the top of the capacitor. The positive lead is on the bottom of the capacitor.



Align the positive lead with the + sign and the negative lead with the white PCB marking.

| Value | Ref | Marking |
|---|---|---|
| 0.47F | C1, C2 | 0.47F 5.5V |

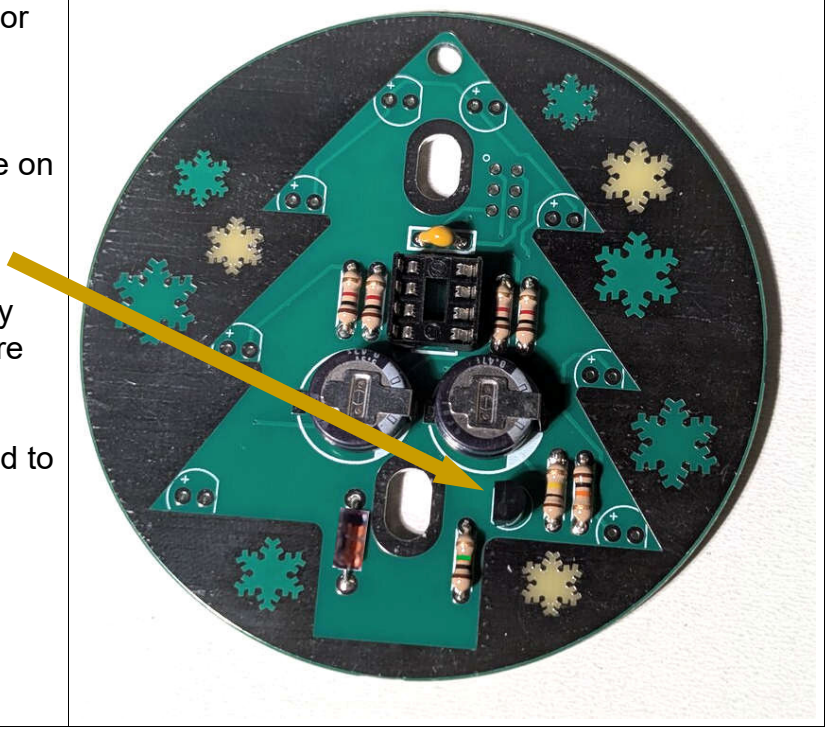
| **Step: 5** | Solder Transistor Q1 |
|---|---|

Solder the 2N3906 PNP transistor Q1 into the area marked Q1.

Ensure that the flat side of the transistor aligns with the flat side on the silk screen.

The pins on this component are quite close together. Ensure very careful soldering so that there are no solder bridges here.

Use a magnifying glass if needed to check the soldering.



| **Step: 6** | Solder LEDs |
|---|---|

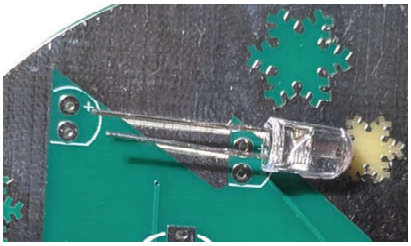
There are 8 LEDs to solder around the edge of the PCB, marked D1 – D8 on the underside of the PCB.

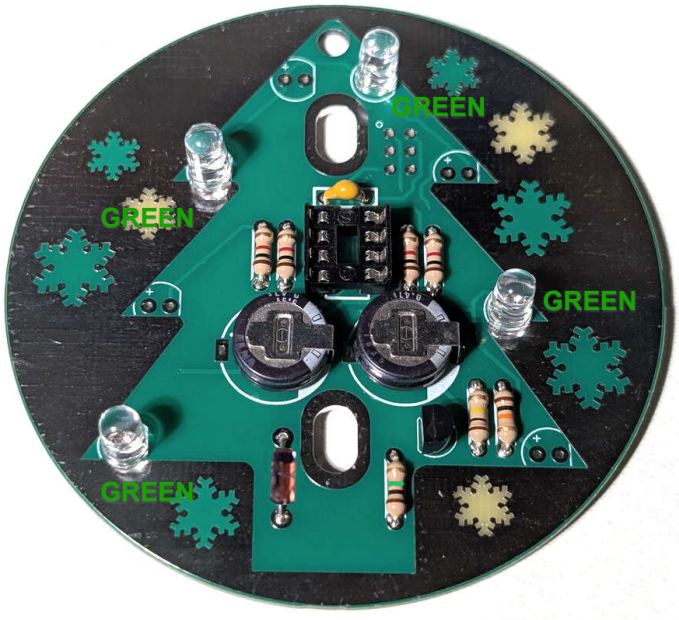
Annoyingly the red and green LEDs are clear and look the same. The RED LEDs will be loose within the kit pack. The GREEN LEDs will be in a small separate bag or on tape together.

Ensure correct orientation!

The **longer lead on the LED is positive**, so place that into the hole marked +. The negative side has a slightly flat edge on the body of the LED.

First solder the GREEN LEDs into positions D1, D3, D5, and D7.



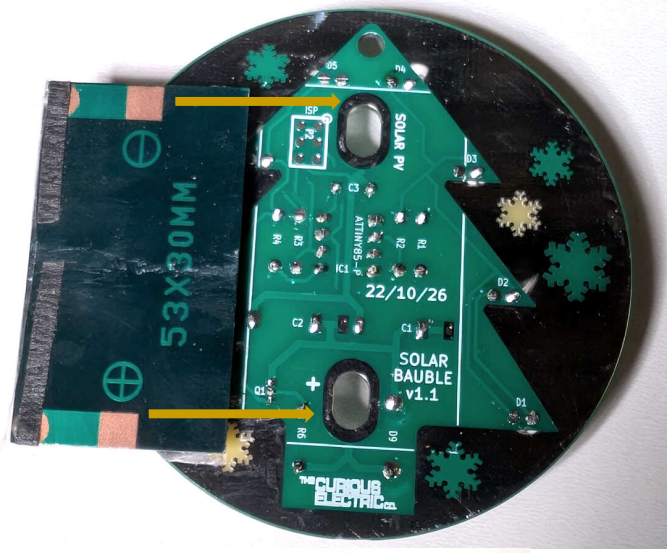


Next solder the RED LEDs into positions D2, D4, D6, D8.

*Note: The LEDs are 'charlieplexed' which is an interesting way of running lots of LEDs from a small number of micro-controller pins.*




| **Step: 7** | Solder Solar Photovoltaic Panel |
|---|---|






Firstly: check the soldering carefully: does it look OK, are there any dry joints? Are there any bridged pads? Also clip off any leads of the components on the back of circuit board. The solar panel will go o the 'back' of the PCB and cover up the soldering we have already done.

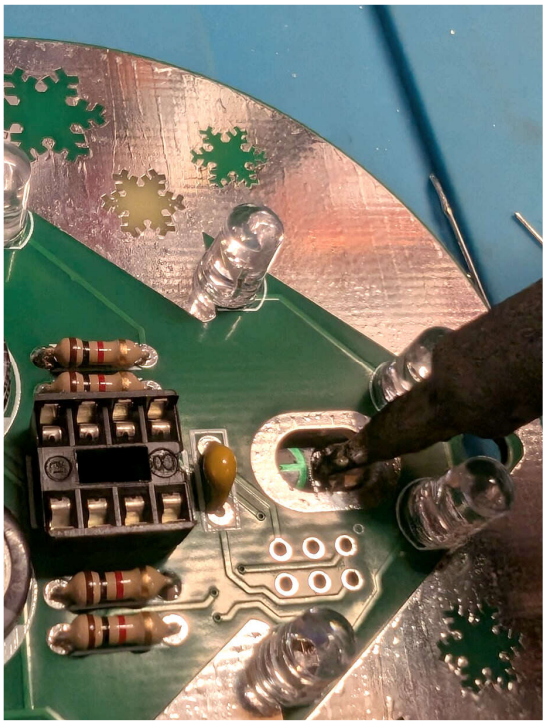
The solar panel must be soldered with the correct orientation.

The solar panel has a + and a – marked side, which must align + to + and – to the other pad (near the ISP and Solar PV writing) on the PCB.

The solar panel is placed flat onto the top of the PCB with the solder pads showing through the large holes.

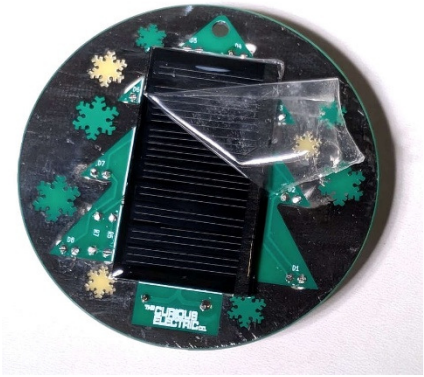
Ensure you put some solder onto the pads of the solar panel. This helps solder the unit to the PCB when we turn it over.

It can help to use some tape or blue-tac to hold it in place here.



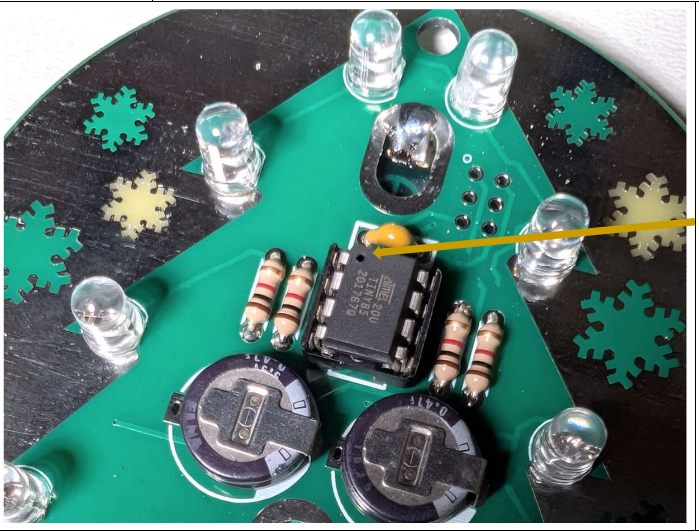
Solder the solar panel pads through the large holes of the PCB.
Getting the angle of the soldering iron right to melt both the solder on the PCB hole and on the Solar panel pad.

You might need quite a bit of solder here!



There is a protective cover on the solar panel. This can be removed after the solar panel is soldered.

| **Step: 8** | Insert the IC |
|---|---|

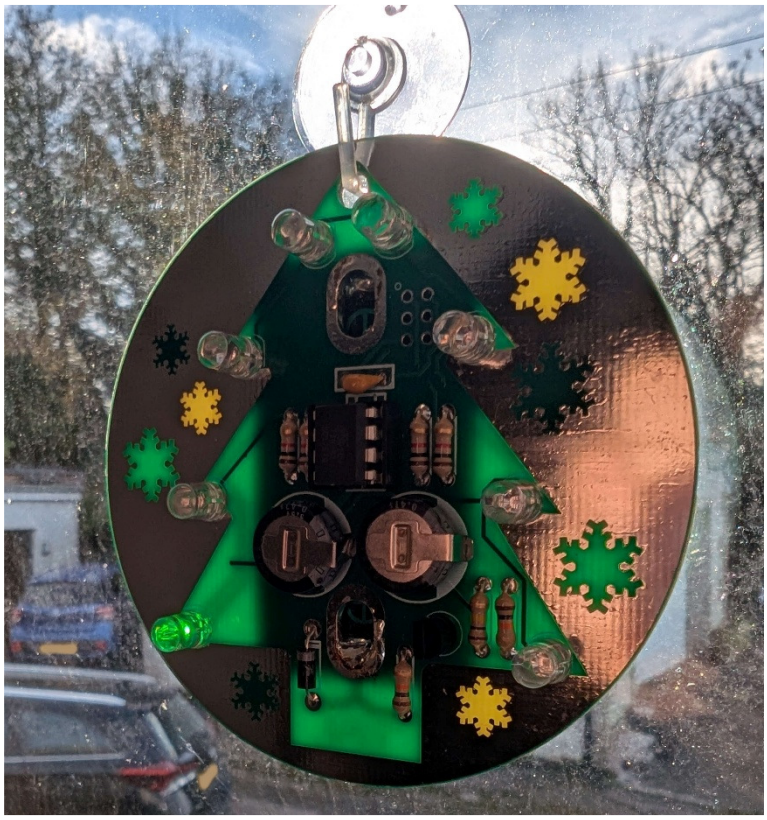


Carefully push the microcontroller into the 8-pin IC holder.

Ensure that the dot on the microcontroller aligns with the notch on the IC holder.

Make sure no pins are bent and they fit into the IC holder.

| **Step: 9** | PCB is finished! Let's test... |
|---|---|



***Please note:*** *This unit is NOT waterproof. Do NOT leave outside if it will get wet.*

Place the finished unit in bright sunlight.

To get a full charge it will need to be left in the sun for a while. It can be left on a windowsill or hung up on a window using the suction hook.

It does need to be placed on a SUNNY window. Preferably south facing or that gets direct sun for some of the day. (It will not charge well from indoor lighting). It will need at least 1-2 hours of sunlight to fully charge.

In sunlight the unit usually will not do anything. *(Note: If in very bright sunlight then the Red and Green LEDs may turn on as this helps protect overcharging of the capacitor.)*

When the solar panel is dark then the Red and Green LEDs should twinkle until the capacitors run down.

The Red and Green will run when capacitors are well charged. When the capacitors drain to a low level, then only the Red LEDs turn on (as they have a lower switch on voltage).

The unit will run until capacitors fully drain and then go to sleep until next charge.

| **Step: 11** | Re-programming Notes |
|---|---|

***Note: No programming is needed – the microcontrollers are already programmed! Here are some notes for people wanting to take things a step further…***

The circuit board has a standard ISP (In-System Programming) port for the ATTiny85 microcontroller and full access to the code which can be programmed via the Arduino IDE, if that's your thing.

You will need an Arduino Uno with wiring set up for programming an ATTiny85 (please search internet).

You will need to include the ATTiny85 programming board manager into your version of the Arduino IDE (please search internet).

You upload code by using the "Arduino as an ISP" option.

This design is open for non-commercial use. A full set of PCB design files, instructions and the firmware is available in the project GitHub repository here:

**https://github.com/curiouselectric/solarBauble**

## Contact details:

We would like you to be happy with this kit. If you are not happy for any reason, then please contact us and we will help to sort it out.

Please email **hello@curiouselectric.co.uk** with any questions or comments.
Please tweet us at **@curiouselectric**

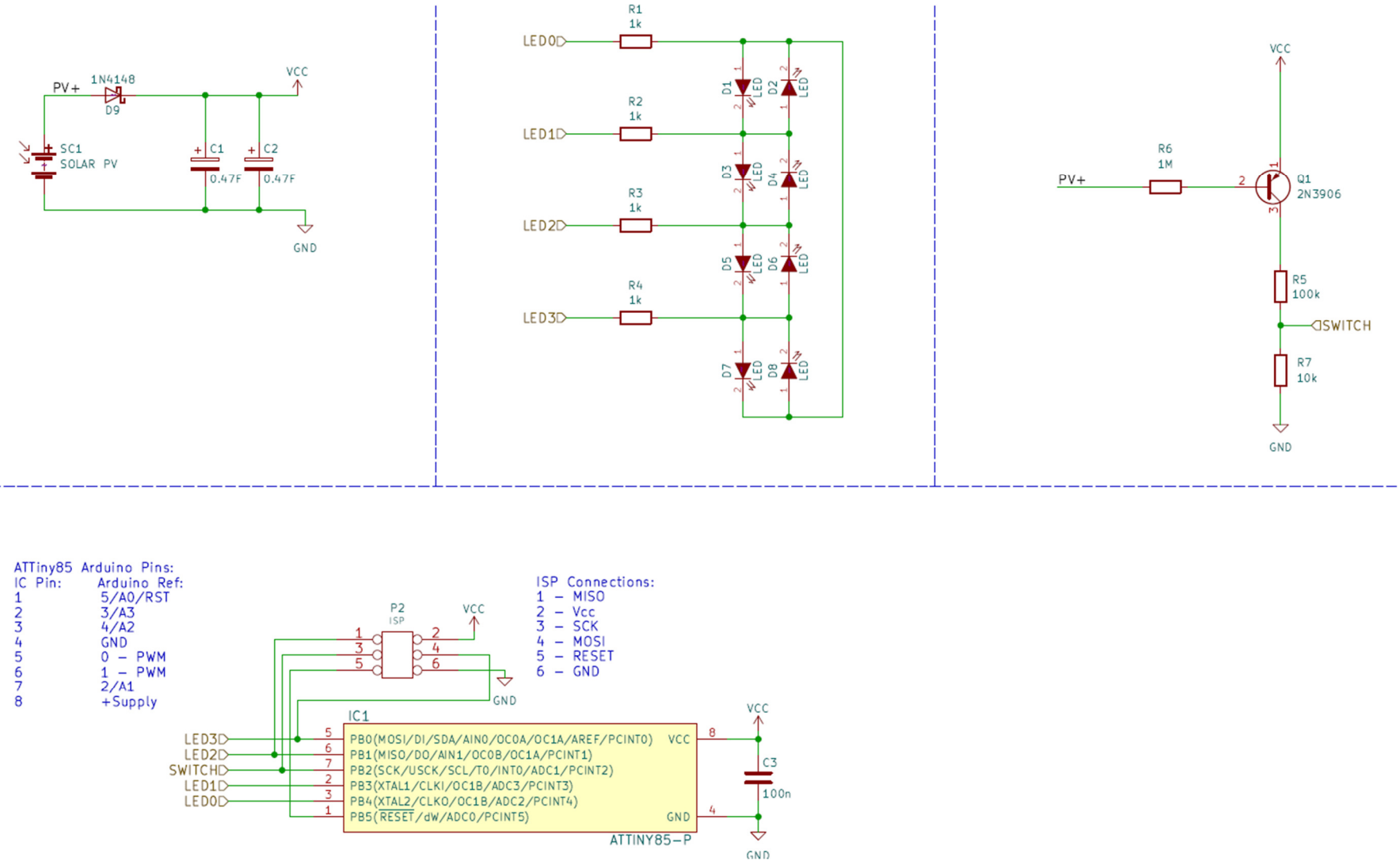
If any parts are missing from your kit, then please email **hello@curiouselectric.co.uk** with details and, if possible, where the kit was purchased.

More technical information can be found via **www.curiouselectric.co.uk**

This kit has been designed and produced by:

**The Curious Electric Company**
**hello@curiouselectric.co.uk**
**www.curiouselectric.co.uk**

# Circuit Schematic




ATTiny85 Arduino Pins:

| IC Pin: | Arduino Ref: |
|---|---|
| 1 | 5/A0/RST |
| 2 | 3/A3 |
| 3 | 4/A2 |
| 4 | GND |
| 5 | 0 – PWM |
| 6 | 1 – PWM |
| 7 | 2/A1 |
| 8 | +Supply |

ISP Connections:
1 – MISO
2 – Vcc
3 – SCK
4 – MOSI
5 – RESET
6 – GND

# Resistor Colour Codes

